Research Methods

Day 9 · Data Analysis — Finding Patterns & Testing Hypotheses

Outline for Today

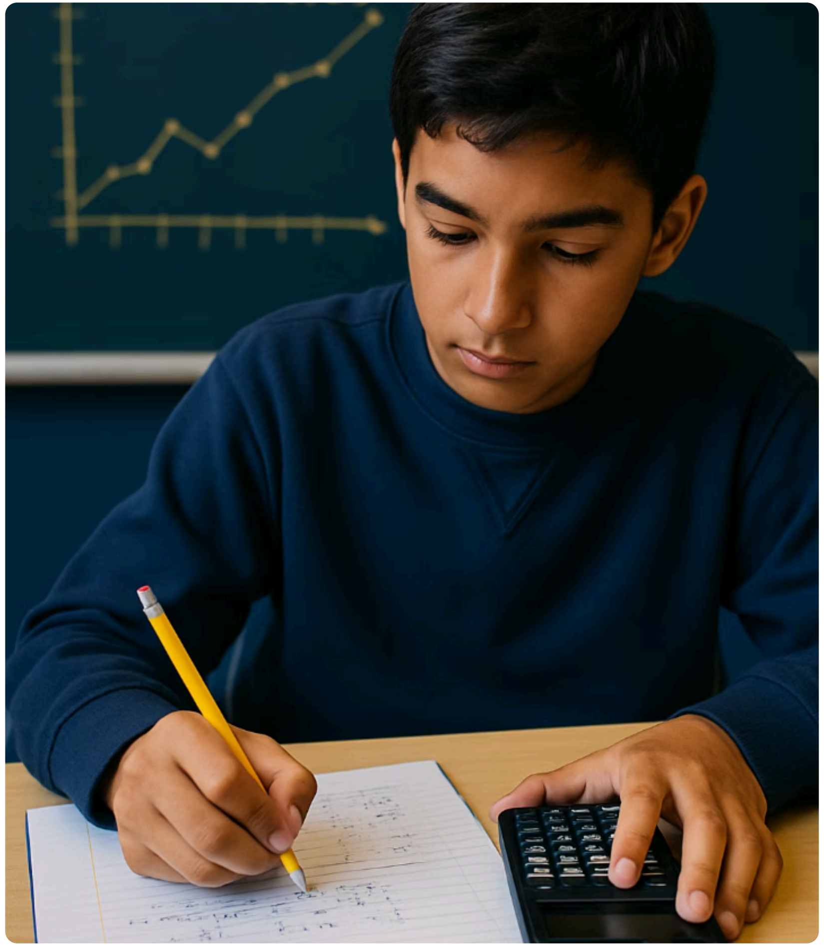
1. **What Does the Data Say vs. What Do We Conclude?**
2. **Finding Patterns, Trends & Outliers**
3. **Correlation, Regression & Causation**
4. **Team Analysis & Instructor Conferences**
5. **Scientific Storytelling: Results Sentences**

Welcome to Day 9!

Today you turn your dataset into **findings** — patterns, relationships, and honest conclusions.

By the end of the day you will be able to:

- Apply **descriptive and basic inferential** statistics to your research question
- Identify **patterns, relationships, and outliers** in data
- Understand what "**supporting a hypothesis**" does — and does **not** — mean
- Connect results back to your **literature review**



What Does the Data Say vs. What Do We Conclude?

Results vs. Interpretation 20 min

The single most important distinction in writing up research:

Results — *"The data showed X."*

What you **observed**. Neutral, factual, numeric. Lives in the **Results** section.

Interpretation — *"This means Y."*

What you **think it means**. Connects to theory and your literature. Lives in the **Discussion**.

Keep them separate. Report the facts first; *then* argue what they mean. Mixing the two is the most common mistake in student research writing.

Practice: Fact or Interpretation?

For each statement, decide: is it a **result** (fact) or an **interpretation** (discussion)?

1. *"62% of surveyed students reported fewer than 7 hours of sleep."*
2. *"Students are exhausted because they spend too much time on social media."*
3. *"Average daily screen time was 4.2 hours, and mean sleep was 6.4 hours."*

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1. ✓ **Result** — a measured fact.
2. 💡 **Interpretation** — a *claim about cause* that goes beyond the data.
3. ✓ **Result** — two measured facts (no claim yet).

Finding Patterns, Trends & Outliers

Descriptive Statistics: A Quick Refresher

Mean

Sum of all values divided by the number of values — the average.

Median

The middle value when the data is ordered from smallest to largest.

Mode

The most frequently occurring value in the dataset.

Range

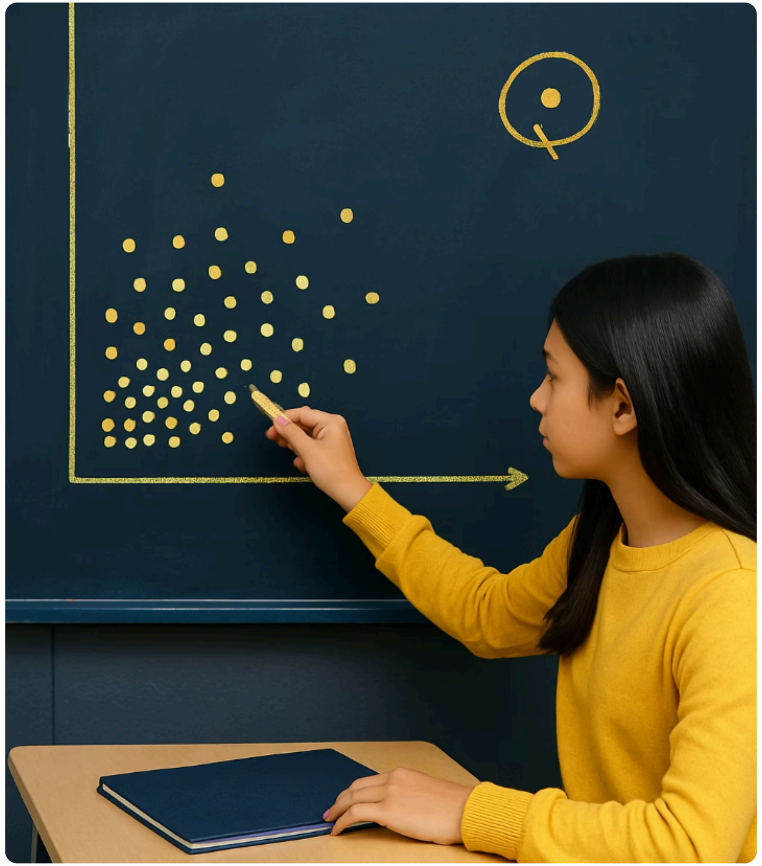
The difference between the largest and smallest values.

Outliers: Values That Stand Out

What are they? Data points much **higher or lower** than most others.

Why they matter: they strongly pull the **mean**, but have little effect on the **median**.

Handle with care: investigate **why** an outlier exists before deciding whether to keep or drop it.



Descriptive Example: Quiz Scores

Dataset: `{4, 7, 8, 9, 9, 10, 10, 10}`

Compute each by hand:

- Mean = ?
- Median = ?
- Mode = ?
- Range = ?
- Possible outlier = ?

Descriptive Example: Quiz Scores

Dataset: `{4, 7, 8, 9, 9, 10, 10, 10}`

Compute each by hand:

- Mean = ?
- Median = ?
- Mode = ?
- Range = ?
- Possible outlier = ?

- **Mean = 8.375**
- **Median = 9**
- **Mode = 10**
- **Range = 6**
- **Possible outlier = 4**

Half the class scored 9+; the most common score was a perfect 10; one student (the **4**) struggled far more than the rest — that's the outlier pulling the mean below the median.

Identifying Trends and Patterns

Trends Over Time

Upward, downward, or flat direction of change.

Patterns in Categories

Consistent relationships between different categories.

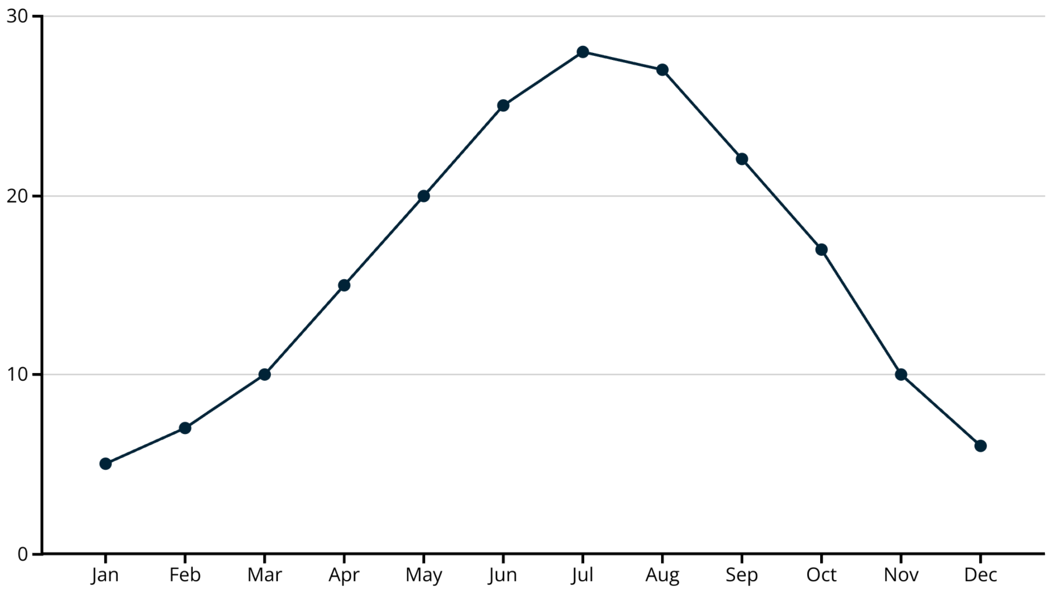
Cycles & Seasonality

Repeating patterns at regular intervals.

Clusters & Gaps

Groups of similar values, or ranges with little/no data.

Trends Over Time

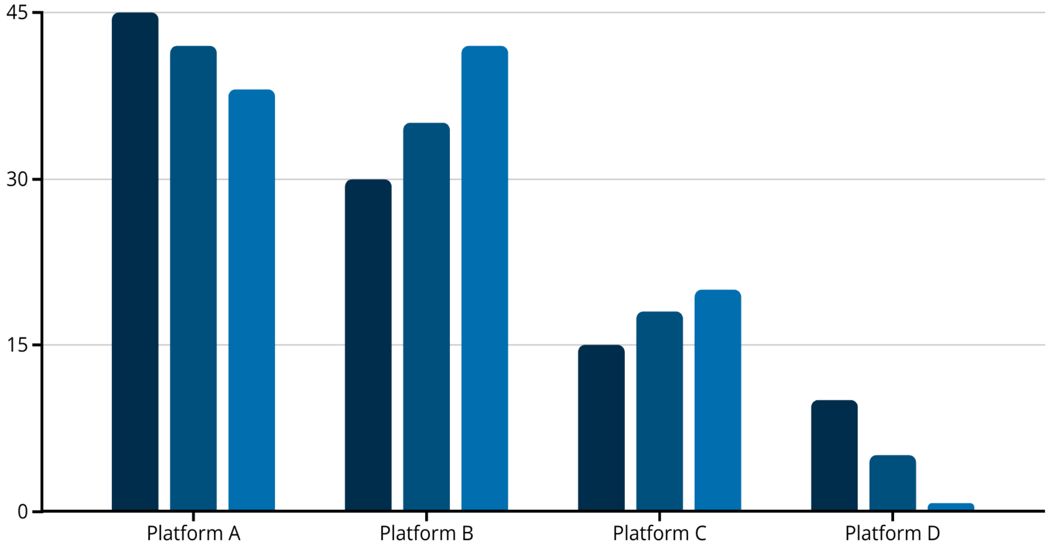


This temperature chart shows a clear **seasonal cycle**:

- Temperatures **rise** in spring
- **Peak** in summer
- **Fall** through autumn and winter

This **cyclical pattern repeats annually** — a trend you'd miss from a single month's number.

Patterns in Categories



Social-media usage across **three years** (2021–2023):

- **Platform A** — declining
- **Platform B** — growing rapidly
- **Platform C** — steady growth
- **Platform D** — disappearing

Comparing categories **over time** reveals direction that a single snapshot hides.

Why Finding Patterns Matters

Make Predictions

Forecast future outcomes based on past patterns.

Inform Decisions

Use data trends to guide choices.

Discover Relationships

Identify connections between variables.

Understand Systems

See how different factors interact.

Correlation, Regression & Causation

Correlation vs. Causation 25 min

🔗 Correlation

Two variables **change together** — as one moves, so does the other.

Example: ice cream sales and sunburns both rise on hot days.

➡ Causation

Changing one variable **directly produces** a change in the other.

Example: pressing the gas pedal makes the car speed up.

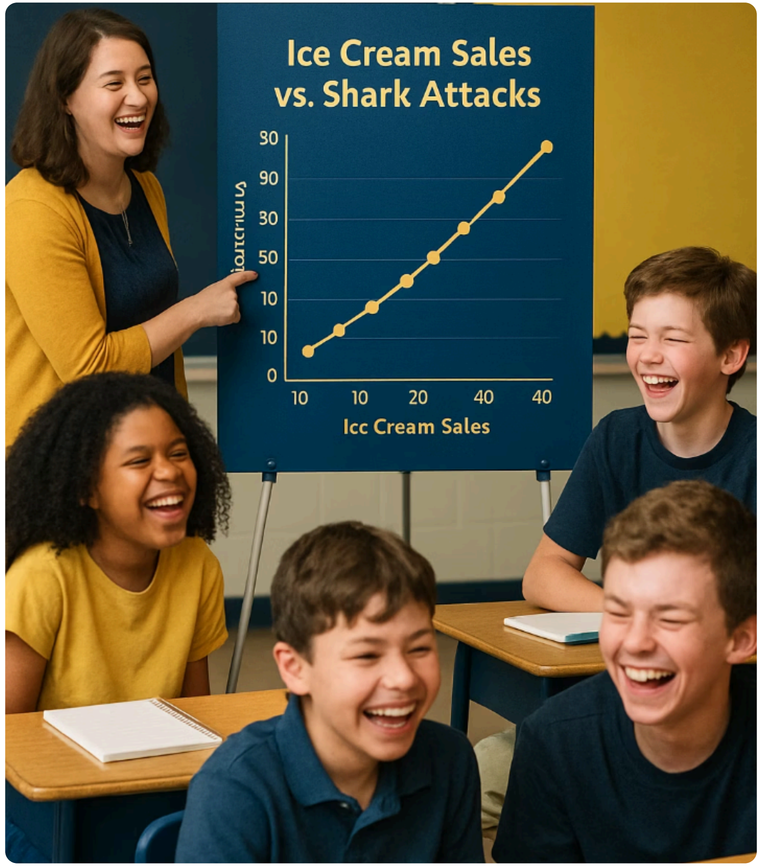
Things happening together does not mean one causes the other. Proving causation usually requires a **controlled experiment**.

Spurious Correlations: Coincidence, Not Cause

Ice cream & shark attacks — both rise in summer, but ice cream doesn't attract sharks.

Backpack weight & math scores — correlated through *grade level*, not because heavy bags teach math.

Hidden variable: often a **third factor** (hot weather, age, study habits) drives *both* variables.



How Do We Establish Causation?

1. **Controlled experiments** — randomly assign subjects to treatment vs. control, holding everything else constant.
2. **Critical thinking** — is there a logical reason X could influence Y?
3. **Reverse causation** — could Y actually be causing X?
4. **Third factors** — could something else drive both?

When *Can* We Make Causal Claims?

The gold standard: a **randomized experiment**. Random assignment makes the groups comparable, so a difference in outcomes can be credited to the treatment.

But observational studies still matter. When experiments are impossible or unethical, careful design and repeated evidence can still build a strong causal case.

Smoking & cancer

Decades of **correlational** evidence — across many studies, populations, and dose-response patterns — established causality *without* a randomized trial.

Natural experiments

Economists exploit policy changes and cutoffs as "found" experiments to estimate causal effects from observational data.

Team Analysis & Instructor Conferences

Team Analysis Time + Conferences 60 min

Work on your **data analysis** while the instructor circulates. Each team gets a **5-minute check-in**.

Your conference, three questions:

1. What have you **found**?
2. What **question** does it raise?
3. What is your **next step**?

Document every finding in your DataCORE journal — even the dead ends. They're part of the story.

60:00

Scientific Storytelling: Results Sentences

Write Three Results Sentences 15 min

Write **three result sentences** about your data — with **precise numbers** and **no interpretation yet.**

1. **A descriptive finding** — e.g. *"The average commute was 27.4 minutes (SD = 8.1)."*
2. **A comparison across groups** — e.g. *"Urban students averaged 1.3 fewer hours of sleep than rural students."*
3. **A relationship between variables** — e.g. *"Screen time and sleep were negatively correlated (r = −0.38)."*

15:00

Homework & After-Class Work

After-Class Team Session ⚙️ 3:15 – 5:15 PM

Teams work with a **TA / peer mentor** available. Produce a **tangible output before dinner** — *not free time.*

1. **Session 9 Mission Sheet (RcF):** visualize one variable, compare two variables, identify outliers, write one key insight.
2. **Session 9 Mission Sheet (RcS):** variable-pair analysis, compute the correlation coefficient, write a 2-sentence interpretation, identify one causal limitation.
3. Create **2–3 publication-quality charts** for your Results slides — clean axes, titles, colorblind-friendly palette, source caption.

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🏁 **Team output due before dinner:** final analysis charts exported as PNG + both mission sheets complete.

Evening Work Period 📘 7:00 – 9:00 PM

Individual + partner work during CTD RTA study hours. TA available for drop-in questions.

- **Individual:** write a **3–5 sentence results paragraph** summarizing your key finding in plain language. Read it aloud — does it make sense **without** the chart?
- **(RcS):** run your **regression / multivariate analysis**. Paste the coefficient table into your notebook and **annotate** what each coefficient means in plain language.

Update your Results slides:

- Insert your **best 2 charts**.
- Add **captions**.
- Make sure the slide **text adds interpretation** — not just a repeat of what the chart shows.